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Biodesign research in the Anthropocene

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Human activity—though recognised as contributing to climate change in the twentieth century works of ecologist William Vogt, conservationist Rachel Carson, biologists Paul and Anne Ehrlich, and philosopher Arne Næss¹—is only now beginning to reverberate within public consciousness as we witness a steep increase in natural disasters reshaping human communities and the planet as a whole. Scientists Paul Crutzen and Eugene Stoermer propose that human activity has had a significant and lasting effect on the natural environment, so much so that this period has been characterised as a new geological epoch: the Anthropocene.² It is in this context that biodesign becomes of increasing interest to designers as we examine ways of making materials that improve the quality of the built environment, encouraging environmental stewardship without exacerbating the ecological problems facing our planet.

Making, as a destructive practice that often requires the consumption of resources, the production of waste and pollution, and the devastation of nonhuman habitats,³ is being transformed through practices that question consumerist culture and look to holistic

ways of employing natural organisms as co-creators. Such practices are in contrast to traditional building practices where products enter a state of gradual degradation as soon as they are complete, before continuing to break down until obsolete.⁴ In nature, living organisms grow and interact with the environment around them. The notion of waste is replaced by continual cycles that ensure the digestion of products through natural processes, a concept that is now being applied within a human context through biodesign.

Through the use of creative practice research, practitioners are able to propose viable alternatives, cultivate an understanding of the ethical challenges that arise as a result of enlisting living biota to solve human problems, as well as facilitate public engagement with issues that are often inaccessible due to the specialist nature of scientific research. The projects I highlight in this chapter seek to differentiate biodesign research from material science and biotechnology by offering a layer of critical thinking that emerges from engaging with the work and methods of other disciplines. The microorganisms are framed as morally considerable actors in their own right, rather than as

Within the built environment, much like the human body, we have assumed an anti-biotic approach that models humanity and cities as independent entities, self-sustaining and impermeable to the wider environment. It is important to tell stories of probiotic worlds when creating more-than-human environments. Being able to conceive of alternative futures must be the precursor to a living building realm.

instruments used to meet human needs, thus proposing a holistic approach. **The creative projects offer an understanding of living organisms as co-creators that are intimately entangled in human life.**

Biodesign, and the stories we tell

Emergent biodesign practices are often associated with technocratic frameworks of engineering and science, where living organisms are frequently conceptualised as a technology that could be enlisted to solve environmental problems and thereby further domesticate nature for capitalist gain.⁵ Efforts to standardise living organisms are often part of a capitalist pursuit to mass produce and increase profit, commodifying living organisms and their functions by establishing further control over nature. However, a more holistic view of living organisms as part of a wider network brings forth questions of the ethical implications of utilising, editing, and cultivating microorganisms for the benefit of humans.

Biodesign practices range from husbandry—encouraging growth and engineering environments that sustain other living species within the built environment—to practices that venture into the realm of synthetic biology—genetically modifying organisms for the purposes of design.⁶ Such endeavours may be necessary in the fight against climate change. However, as designers, we have a responsibility to foster conversations about considerations beyond the needs of human society when undertaking biodesign explorations. As we venture into a new era in material development, conversations surrounding the nature of

working with living biota become important in shaping societal attitudes and in challenging the social structures that have permitted the exploitation of the natural environment.

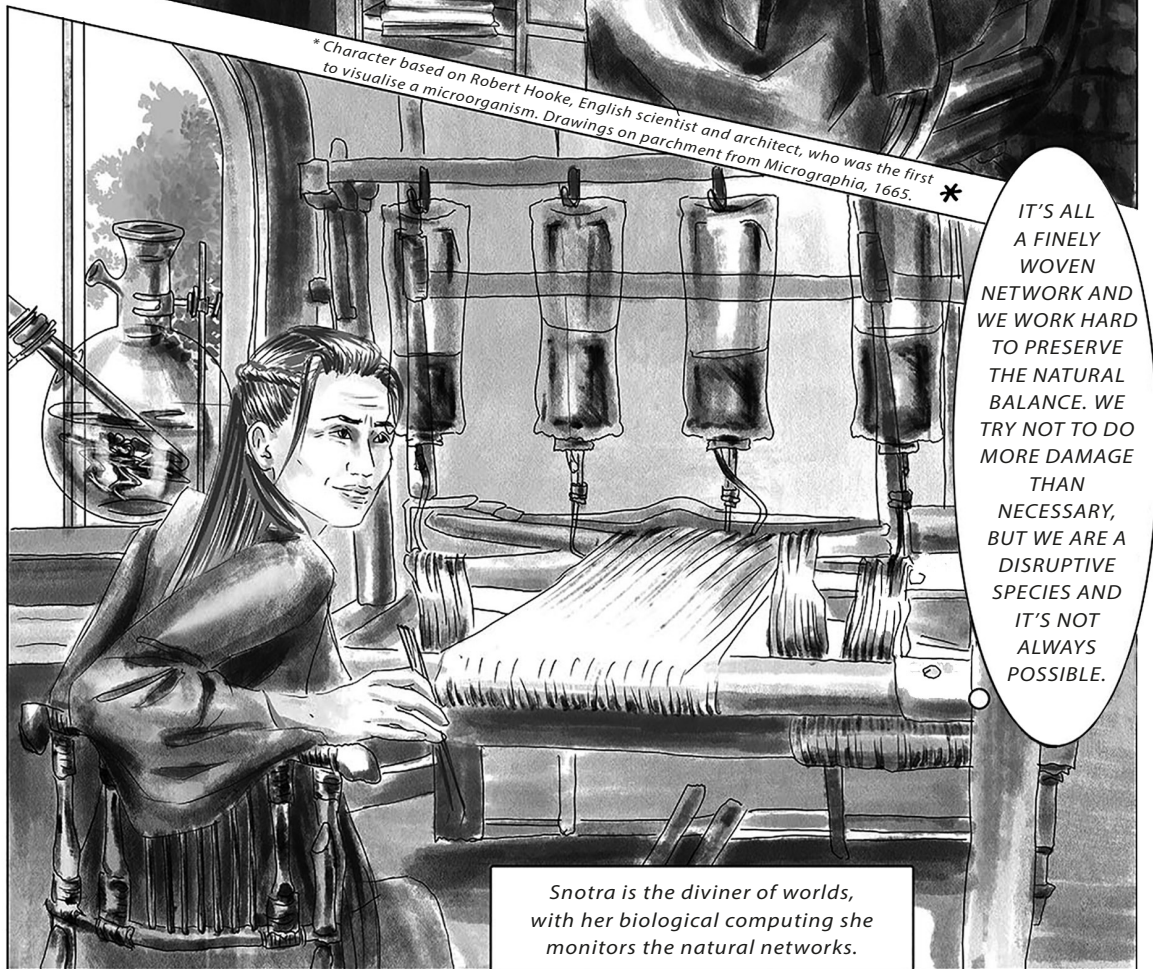
To develop futures that prioritise the planet's health, I look to bridge scientific practice and certain indigenous understandings of humanity's place in nature. The spoken tradition and narrative development within key indigenous cultures has been used as a tool to govern human behaviour and prevent inadvertent damage.⁷ **Therefore, narrative creation helps redefine the technocratic frameworks present within biodesign practice by postulating alternative futures that look to indigenous knowledges and relationships with nature, fostering environmental thinking beyond formal institutional frameworks and Western, anthropocentric definitions of nature.** The narratives that emerge seek to empower individuals and assign responsibility back to practitioners, encouraging behaviours and reminding the public of their agency to incite change. As opposed to the individualism that has permeated contemporary consumer society,⁸ it is an effort to encourage the co-operative effort of individuals as active participants in society.

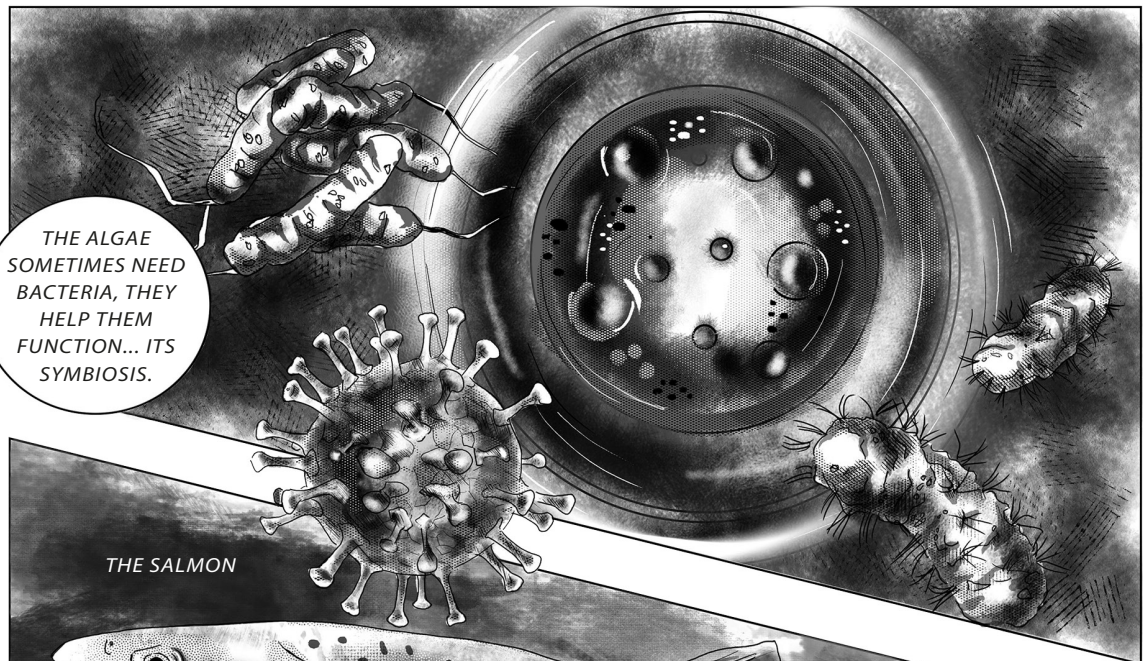
The graphic novel *Bios in Search of Zoë* (unpublished, 2021) captures conversations that occur within a scientific setting and juxtaposes them with indigenous knowledge that is often marginalised within Western discourse (Figures 17.1–17.3). The work builds upon an existing

17.1 – 17.3. (Overleaf) Excerpts from graphic novel *Bios in Search of Zoë*, an eco-fictional narrative.
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* Character based on Robert Hooke, English scientist and architect, who was the first to visualise a microorganism. Drawings on parchment from *Micrographia*, 1665. *

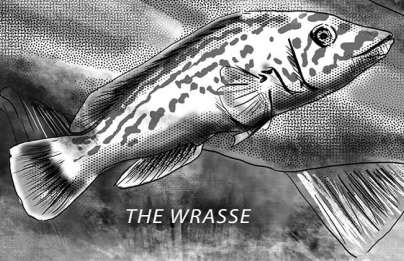
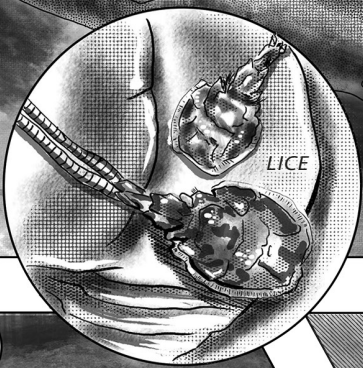




THE SALMON

IT'S ALL CONNECTED, MAN!

IT'S LIKE WHEN YOU FARM SALMON, YOU NEED THE WRASSE TO FEED ON THE LICE THAT FEED ON THE SALMON.



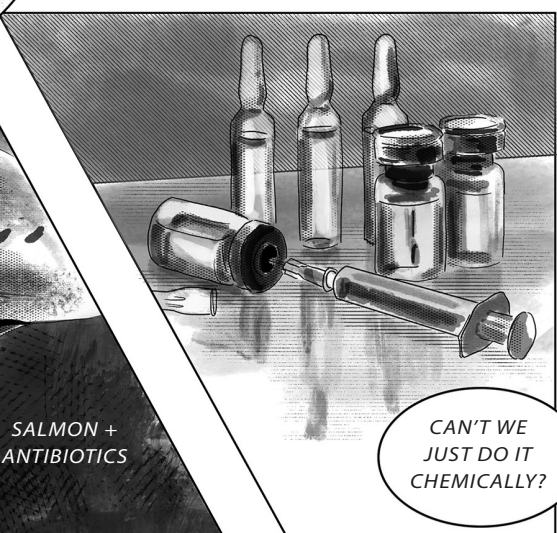
THE WRASSE



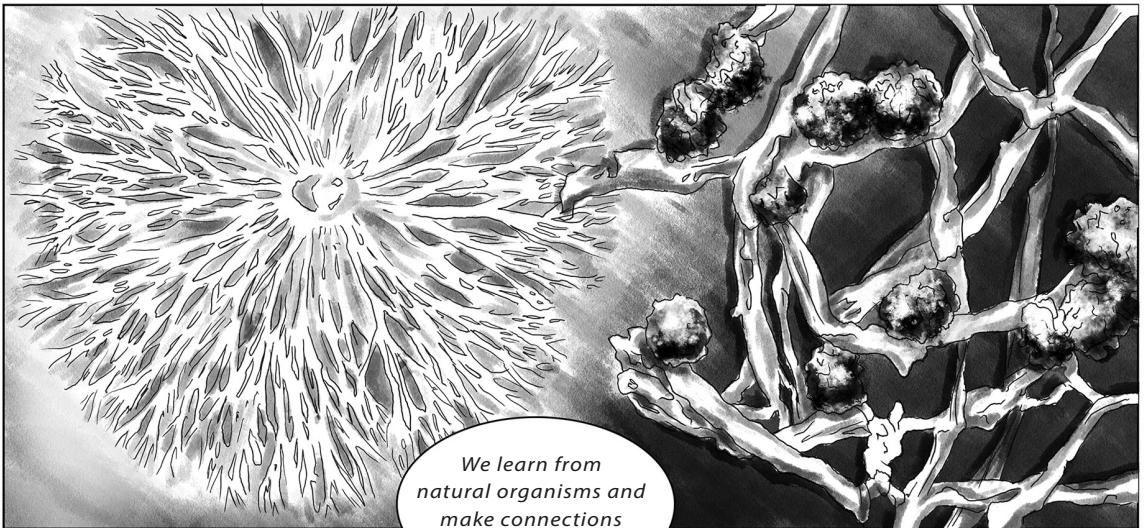
WE TRIED THAT...

IT DIDN'T WORK OUT.

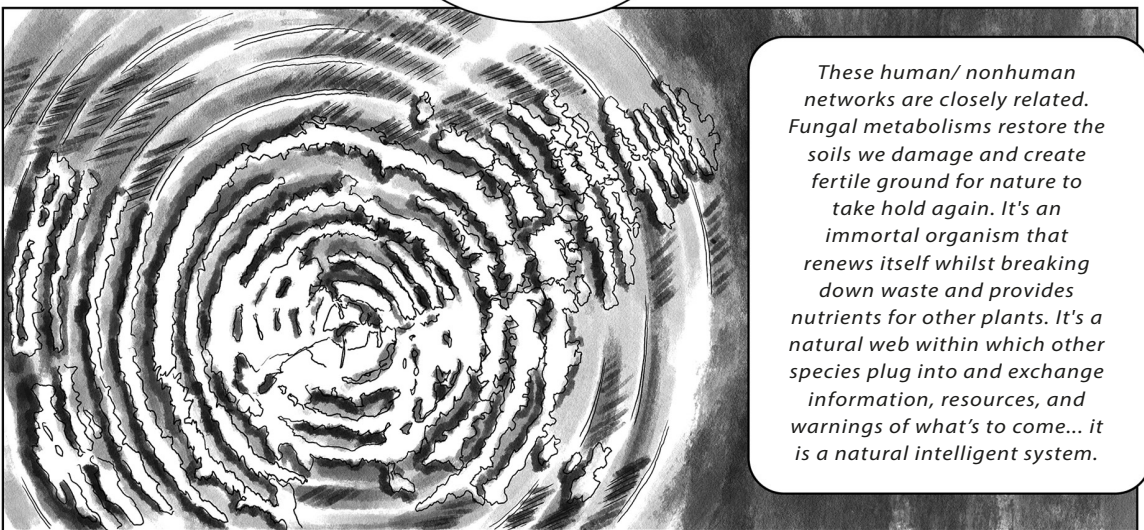
SALMON + ANTIBIOTICS



CAN'T WE JUST DO IT CHEMICALLY?



*We learn from
natural organisms and
make connections
between scales.*



*These human/ nonhuman
networks are closely related.
Fungal metabolisms restore the
soils we damage and create
fertile ground for nature to
take hold again. It's an
immortal organism that
renews itself whilst breaking
down waste and provides
nutrients for other plants. It's a
natural web within which other
species plug into and exchange
information, resources, and
warnings of what's to come... it
is a natural intelligent system.*



*We try to
work with these
organisms to
create what we
need.*

*If a tree struggles
due to lack of
sunlight, these
underground networks
can redistribute
nutrients to help its
survival.*

strand of ecological narrative creation explored by feminist scholar Donna Haraway, living architecture pioneer Rachel Armstrong, and authors Ursula le Guin and Octavia Butler.⁹ The narrative proposes an alternative societal structure that uses animism in conjunction with contemporary scientific knowledge to foster a human existence that does not disrupt natural networks. Scholars such as David Graeber surmise that our reluctance to imagine futures outside the existing framework is because capitalism has successfully stifled our ability to do so.¹⁰ The novel seeks to connect emerging research into animal sentience and plant intelligence with the practices of indigenous cultures that assign agency to nonhuman organisms and the planet, and by doing so discourage the abuse of resources. In this example, the creation of the graphic novel becomes a tool for exploring the necessary changes that must occur so as to design for a holistic built environment.

Bottom-up and top-down: turning architecture on its head

To aid uniformity and accommodate contemporary building processes of standardisation and mass production, architects often think of a kit of parts.¹¹ It is a form of thinking that closely resembles systems engineering and is predicated on control and the choreography of static elements. Within biodesign, however, notions of inertia become problematised, since nature presents a vast array of active processes that do not function in isolation and that are both inward orientated, sustaining organismal health, and outward sensing, responding to environmental

conditions.¹² The process of working with living things becomes a give-and-take of getting to know the behaviours and needs of a living co-creator, as well as the networks that it forms and relies upon.

Here, we venture into ecological thinking, where isolated *in vitro* settings often used within controlled testing are substituted with a multi-variable environment. There, the organisms fight to survive and impact upon the wider ecology by competing with other species, releasing by-products affecting and altering the environment around them.

We have the choice of engaging in biodesign through the curation of proposals for the integration of living biological processes within our built environments, establishing desired outcome goals, and structuring sustainable material lifecycles. This is often referred to as a top-down approach and is particularly effective when bringing living organisms into artificial human habitats. In the top-down approach, the designer resides closer to the realm of controlled, solution-driven design, where the outcome is envisaged before the process allowing it to manifest has unfolded. In this mode of working, there is a need to define clearly the role of the organism as more-than-instrumental, otherwise there is a risk of falling into a technocratic mode of objectifying non-human life. It becomes tempting to revert to building block analogies that can be selectively employed in a plug-and-play system that seeks to simplify organismal complexities.

At the other end of the spectrum, the bottom-up approach to biodesign is based on a close understanding

of the organisms and originates in identifying organismal behaviour and environmental needs. Brought into the building realm through the construction of nonhuman support systems, other species are then allowed to proliferate our historically exclusionary built environment. Within biodesign, the bottom-up approach opens-up conversations about co-habiting with microorganisms and creating probiotic human environments. It is in this context that we begin to imagine living surfaces that accommodate biofilms that enrich the quality of living spaces.¹³

The role of the architect shifts, here, from designing for humans to designing for other living species, a practice that requires a close relationship with the organisms we seek to accommodate and that recognises the nonhuman as creative entities. It further enforces the fact that the response of these living organisms matters and an understanding of the ways they experience the environment is essential in this co-creative endeavour. For designers, this often entails expanding our skills and knowledge so that we can actively participate in early-stage laboratory investigations.¹⁴ An example of such work is the collaborative development of photosynthetic bio-composite ceramics that formed a series of fundamental studies as part of my creative practice research (Figure 17.4). Bio-composite development becomes an exercise in symbiopolitics, defined by cultural anthropologist Stefan Helmreich as the managing of symbiotic



17.4 Photosynthetic bio-composite. © Author in collaboration with Gary Caldwell's Lab, School of Natural and Environmental Sciences, Newcastle University, 2021



relationships to achieve desired environmental goals.¹⁵ The structures that emerge postulate surfaces that are living, breathing, teeming with life, visually changing and interacting with the microbiome of human habitats. The process of developing this new type of living building fabric is intrinsically linked to repositioning humanity as an actor in a nature that is no longer subservient to human needs.

Holobionts, contamination, interaction, and creatively managing life

Within the built environment, much like the human body, we have assumed an anti-biotic approach that models humanity and cities as independent entities, self-sustaining and impermeable to the wider environment. Such views of hygiene have served as a model for health and wellbeing throughout modernity.¹⁶ However, such neo-Darwinian views of individual competition¹⁷ are coming under scrutiny with research into the human microbiome, which reframes our understanding of humans as holobionts sustained through a myriad of symbiotic relationships with microorganisms that regulate and perform many vital functions within the human body.¹⁸ Similarly, the built environment within developed countries has been guided for the better part of the twentieth century by dysbiotic interpretations of hygiene that have created human habitats that no longer provide exposure to friendly symbionts. These reductionist views of hygiene within architecture are currently being

*17.5 3D-printed ceramic pieces, Mitochondrial Matrix.
© Author in collaboration with Wellcome Trust Centre
for Mitochondrial Research, Newcastle University, 2021*

challenged by research into microbial environments. Such work looks at the wider impact of microbial diversity in human habitats as demonstrated in the integration of biomaterials within domestic settings. Furthermore, speculative design work that ventures beyond material development is also a valuable tool in expanding traditional definitions.

The view of humans as holobionts helps reposition humanity as dependant on nature rather than an independent actor outside of natural networks. Evolutionary theorist Lynn Margulis speculates that mitochondria found in eukaryotic cells originated as alpha proteobacteria that two-and-a-half million years ago joined with hyperthermophilic archaea.¹⁹ This symbiotic relationship allowed mitochondria to survive the oxygen-rich environment and provided the majority of eukaryotic cells with the energy needed for respiration.²⁰ The relationship between mitochondria and the human body became the focus of an interdisciplinary project with the Wellcome Trust Centre for Mitochondrial Research, Newcastle University (Figure 17.5). Practitioners from the fields of art, architecture, and creative writing were paired with scientists investigating mitochondrial disease. The early conversations served as a catalyst for creative work that aimed to overcome the barrier of jargon-laden scientific knowledge to communicate to the public the complexities of the disease. This type of research through conversation highlights the role that language plays in shaping our relationship with nature.

The piece that emerged from this collaborative work aimed to capture the

variations and anomalies in the genetic code that are experienced by patients with an array of symptoms affecting everyday life. Entitled *Mitochondrial Matrix*, the work consisted of three hundred three-dimensional printed ceramic vessels that were based on simple subdivision and that gradually increased in complexity to illustrate the anomalies within the genetic code. The pieces were individually glazed, thereby making duplicates that nevertheless appeared visually distinct from one another, and the overall composition formed a spiral, beginning with simple geometries before rippling out into distorted and grotesque morphologies. The choice of ceramics at a human-scale helped to bridge the gap between the micro-scale of the human cell and the patient experience. The pieces were often anthropomorphised by the viewer, with a desire to hold and touch the individual vessels arising, enforcing a physicality akin to physical manifestations of the disease on a human body-scale.

Something that emerged with the scientific partners was the recognition of the processes they had described within the ceramic installation. Although not biologically accurate, the expression of the complexity observed within the medical research conducted fostered recognition within the observer. In this instance, the biodesigner acts as mediator between the inaccessible world of science and the public. This type of creative practice deploys skills more closely grounded in design to engage in questions of human/nonhuman relationships and relate scientific research to our understanding of the world.

Shaping worlds

Our current reluctance to envisage futures beyond the existing status quo, in particular the capitalist reality to which we struggle to imagine a viable alternative, is deeply rooted in the stories constructed by modernity. Therefore, the stories we allow ourselves to tell of possible futures are in line with capitalist growth principles and ever-growing human demands for comfort through consumption. This yearning is closely linked to our innate need for security met by the constant environmental conditions that have allowed humanity to populate, domesticate, and above all commercialise most of the planet. As we reach a tipping point that brings us to the end of the environmental, and therefore economic, stability that our species has recently enjoyed, we look for ways to survive in a volatile landscape. Our ability to imagine futures other than the ones on the consumer capitalist trajectory becomes a means of reconciling humanity with a nature that has been disturbed and altered and that is laden with uncertainty. It is this uncertainty that calls for the construction of viable futures which advocate a societal shift to new modes of co-existence.

Stories of more-than-human environments can adopt many narrative forms that become tools for testing hypotheses and for understanding issues on a more intimate level. The fictional work, *Bios in Search of Zoë*, served as a speculative scenario-based testing tool that wove together the practical application and wide-scale adoption of biotechnology within society, along with the requirement for a societal shift that would facilitate

necessary compromises in lifestyle.

Although the narrative takes the form of fantasy fiction, it indirectly points to the question of whether progress can exist without growth, an idea explored by anthropologist Anna Tsing in her search for ways to exist in capitalist ruin.²¹ However, the demonstrated practices also highlight our close entanglement with nature, our fragility and dependence, as well as our ability to alter and co-create.

A fine balancing act occurs when working with living organisms. As designers we aspire to propose solutions and foster meaningful conversations, however the tools that we employ require careful examination to prevent a perpetuation of exploitative practices. The works presented highlight the importance of interdisciplinary collaboration within creative biodesign practice and suggest an ethical responsibility for designers to engage with emerging research in a way that fosters a healthier relationship with nature that challenges technocratic notions of life. Rather than further subjugating the natural environment, biodesign can become a tool for reframing the way nature is viewed. The purpose of the creative practice work presented is, therefore, to rekindle a reverence for nonhuman life that has been systematically stamped out since the Enlightenment.

Acknowledgements

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